Named Entity Recognition for Weather Domain in the Marathi Language

Tarun Tapadiya
Department of Computer Engineering
COEP Technological University
Pune, India
tapadiyatr19.comp@coep.ac.in

Sayali Patil
Department of Computer Engineering
COEP Technological University
Pune, India
patilsv19.comp@coep.ac.in

Vedant Bairagi
Department of Computer Engineering
COEP Technological University
Pune, India
bairagivg19.comp@coep.ac.in

Dr. P. R. Deshmukh
Department of Computer Engineering
COEP Technological University
Pune, India
dpr.comp@coep.ac.in

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***Abstract*—A vast volume of content is being published on the internet every second as a consequence of the advancement of digital media and the growing breadth of journalism. Computationally, this unstructured data is opaque. Named Entity Recognition (NER) is a two-step process of finding and then classifying crucial textual data. Most Named Entity Recognition work has already been addressed in English literature compared to other Indian languages. There are restrictions on research in the Marathi language because of scarce resources, and regional and morphologically complicated language. This work is concerned with the Weather Entity Recognition framework in the Marathi language. For predicting the sequences, Conditional Random fields(CRFs) and bidirectional LSTM (BiLSTM) models are used. To represent the word vectors, fastText and Transformer based mBERT embeddings are used. The result observed in the Bi-LSTM CRF model with FastText word embedding has an accuracy of 97.45%.**

***Index Terms*—Name Entity Recognition-NER, Marathi Dataset, word embeddings**

I. Introduction

The information age of today is accelerating astonishing advancements at a breakneck pace. Information is changing due to the increasing adoption of technology, AI, and cloud development techniques. With improvements in digital platforms and electronic media, the growth of digital content is accelerating at never-before-seen rates. Almost all news organizations update every occurrence in real time via their web platforms in the form of articles, blogs, manuscripts, and a myriad of other formats. The data produced doesn't have a uniform format. The data is divided into Structured, Semi-structured, and Unstructured formats. For unstructured data, computational complexity and machine understanding of the data is very high. Computationally, this unstructured data is opaque. Finding a targeted event and its arguments manually in such a vast ocean of information is a very difficult operation. Identifying data and chunking entities is made easier by Named Entity Recognition (NER). Named Entity Recognition (NER) is an important branch of Natural Language Processing (NLP). Information Extraction includes the task of Named Entity Recognition (NER). Named Entity Recognition (NER) has already been the subject of in-depth research all across the world. Since English is the most widely spoken language worldwide, more NER work has been addressed in English literature than in other Indian languages. The majority of online content on the internet is in Indian languages. India's central and western regions are home to the Indo-Aryan language Marathi. Marathi is the second official language in Goa and is mostly spoken by residents of the Indian state of Maharashtra. Limitations being created in research because of Ambiguities, Abbreviations and non-local dependencies, Agglutinative and inflectional nature and many more issues as discussed in [1].

While various sectors use weather data for real business cases like travel planning, demand forecasting, and supply chain management, this study is concerned with the Marathi language version of the Weather Entity Recognition framework. Weather warnings protect human life and property, this is the reason why they are important forecasts. In this work, key information from the weather domain is identified and then categorized in Marathi using predefined categories.

There are majorly three approaches used for the classification of Named Entities into predefined classes. They are defined in TABLE I.

The Named Entity Recognition system built on a supervised machine learning (ML) method is described in this study. The paper suggests using a BiLSTM and conditional random fields

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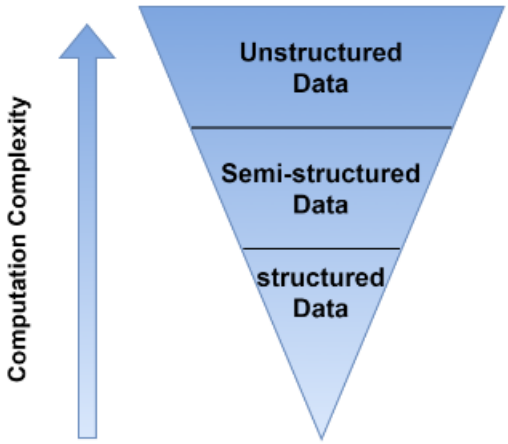

Fig. 1: Types of Data

(CRF) neural network together with word vector fastText, and transformer-based mBERT embeddings for the Marathi text.

TABLE I: Different Methodologies for NER

Technique	Features
List Lookup Approach	• Gazetteers, dictionaries, or lexicons are used • Contains a collection of vocabulary • Basic string matching algorithms are used • Has limitations of regular updation and maintenance
Rule-based Approach	• Utilises a set of custom rules • Rules are based on the pattern in the text and syntactic structure. • Focuses on tokens and describing them with features • For morphologically rich and agglutinative languages it fails to produce the required results.
Machine Learning Approach	• Make use of Mathematical Logic and formulas • It is a corpus-based approach • Corpus is a large set of annotated data • Minimises manual efforts and can be portable to another domain.

II. Literature Survey

The Markov Chain Property is the foundation of the Hidden Markov Model (HMM). In this generative model, states are hidden, the likelihood that the next state will occur depends only on the probability that the previous state will occur, and future states are not taken into account. In [2], Gayen and Sarkar have created an HMM model on a dataset that is tagged in IOB format. They have developed a Named Entity Recognizer for 7 Indian languages. For evaluation, they considered traditional evaluation matrices. For the Bengali language, they achieved the highest F-Measure of 85.99%. They have achieved an F-measure of 42.89% for the Marathi Language.

In [3], authors have used HMM approach and got an accuracy of 90% using a rich corpus. Stacked bigram HMM with Katz Backoff smoothing is implemented by [4] and an overall performance of 75 % is achieved. Authors have achieved an F1-Score of 97.14% for Hindi using HMM as discussed in [5]. In [6], a Trigram HMM with Laplace smoothing is used and an F1-Score of 72% is achieved.

Conditional Random Fields (CRFs) take contextual information into consideration. It is a discriminative model used for predictions. In [7], they have developed a statistical CRF model to be used with the data, which consisted of 27,177 Marathi sentences with 63,236 distinct words. Authors have achieved F1-Measure of 75.7% and 75.51% for overall Named Entity(NE) identification and for average Named Entity(NE) classification respectively by using this probabilistic finite state automata-based Mner-CRF model.

In [8], Sequential data is labelled using CRF++, which is an open-source version of Conditional Random Fields. Contextual word characteristics of 3, 5, and 7-word word windows have been applied. In comparison to Word Windows 3, it has been shown that Word Windows 7 and 5 have produced greater outcomes. Moreover, for various training sets, the Recall value of word window 7 is marginally lower than that of word window 5, while the Precision value of word window 7 is greater than that of the 5-word window.

In [9], authors have discussed a hybrid model for identifying and classifying named entities, by combining a Rule-based model with a Statistical model. They have created a Hybrid architecture by using HMM system and handcrafted rules, a list of gazetteers. By dividing the corpus into two sets with an 80:20 and 90:10 percentage proportion, the dataset is partitioned into two sets one for training and another for testing purposes. In Marathi, the system can identify 12 types of named entities with an average accuracy of 89.05% and 90.09% for NE identification and NE classification respectively.

Neural networks are being used in recent years due to the increase in computational power and their ability to generalize. Layers and filters are building blocks of a convolutional neural network, with the Input layer being the entrance to the model, Output layer being the exit of the model. The hidden layer is the main component of CNN. Convolutional neural networks (CNN) with the help of convolutional layers apply filters to input for activation of results. The components of a convolutional neural network are discussed in [10]. In [11], authors have given a 300-dimensional word embedding vector as input to 1D convolution. The authors have achieved F1-Score of 72.2%, 70%, and 73.7% for deep learning models CNN, LSTM, and biLSTM resp., which are combined with transformers.

In [12], Rule-based NER architecture is applied to the Malay language. Tokenized words are first evaluated using a

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POS tag dictionary. Then tokens with proper noun tags are classified into named entity classes according to prefixes, suffixes, indicator words, and different rules to identify person, location, and organization.

A biLSTM model is an advancement over the unidirectional LSTM model. It is a sequence processing model that runs input in two ways (forward and backward) [13]. In [14], authors have discussed biLSTM with GloVe embeddings and achieved an accuracy of 90.32%. In [15], the NER system for the Hindi language is developed by using ELMo (Embeddings from Language Models) with BiLSTM neural network. The embedding layer propagates context learned from the named entities to two stacked Bi-LSTM layers for classification. They got 98.67% of average accuracy.

Transformers [16] have become state-of-the-art in the field of NLP. In [17], various transformer architectures are used after fine-tuning of their corpus. Of which MahaRoBERTa gives the best F1-score, 85.30%. mBERT gives best performance with F1-Score of 91.27% for Marathi NER while IndicBERT [18], a part of the IndicNLPSuite gives F1-Score of 88.71% .

III. Proposed Methodology

A. Dataset

The dataset consists of 2517 Sentences. For the dataset 300 Marathi news articles were scraped from lokmat.com using Selenium and the requests library. The dataset contains extremely few English words and almost entirely Marathi sentences. Chosen newspaper articles for the dataset don't include extraneous details like publication authors, publication profiles, etc.

For tokenization, indic_nlp_library was used. After cleaning the dataset manually and preprocessing the dataset statistics are as follows:

- Number of tokens: 34775
- Number of sentences: 2517
- Number of tags: 17

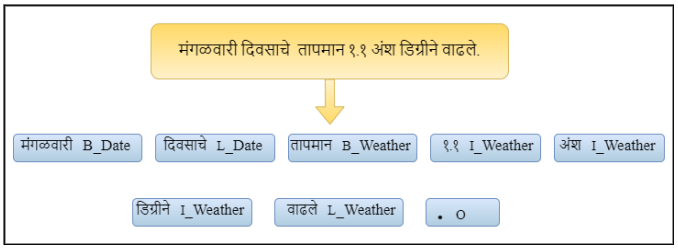

Fig. 2: CoNLL BILOU scheme

The complete dataset is manually tagged with four named entity classes. These classes include Weather, Location, Date, and Organisation. Annotations are done using the CoNLL BILOU scheme in which B_X tag indicates the beginning word of X entity, I_X tag indicates the inside word of X entity, L_X tag indicates the last word of X entity tag, U_X tag indicates a single word of X entity and O for other tag or not a named entity. The work introduces a novel dataset for NER, providing a valuable resource for the development and evaluation of NER models in weather domain.

TABLE II: Dataset Description

Tags	Description
B_Weather	Beginning word weather
I_Weather	Inside word weather
L_Weather	Last word weather
U_Weather	weather (One word)
B_Loc	Beginning word Location
I_Loc	Inside word Location
L_Loc	Last word Location
U_Loc	Location (One word)
B_Org	Beginning word Organization
I_Org	Inside word Organization
L_Org	Last word Organization
U_Org	Organization (One word)
B_Date	Beginning word Date
I_Date	Inside word Date
L_Date	Last word Date
U_Date	Date (One word)
O	Other

B. Word Embeddings

Compared to textual data, computers are better at handling numerical data. Hence, text embeddings aid computers in processing language in the same way that people do. There are primarily two methods for embedding text: either capturing word semantics in context, as opposed to other embeddings (Contextual Embedding), or capturing latent syntactic and semantic properties (Classical Embedding). Both Classical and Contextual Word Embeddings are considered for the NER work. Fasttext and mBERT embeddings are picked for this work because of the Marathi language's morphological intricacy. The category of Classical word embedding includes FastText embedding. A neural language model is used by FastText embedding to learn word representations. Instead of treating a word as a whole, it embeds words by dividing them into several sub-words (or n-grams) and then putting those into the neural network. It is capable of learning words that are new, uncommon, and foreign [19]. Multilingual BERT, also known as mBERT, is used for contextual word embedding. It was developed in a number of Indian languages, including Marathi, and is essentially a deep learning model. It considers the contextual meaning of words and Depending on the context of the sentence portrays each word in a different way.

C. Preprocessing

Sentences that are composed of the word and tag pairing make up this dataset. In order to provide them with indexing, the number of distinct terms and tags across the entire dataset is calculated first. Word embedding vectors are generated by iterating through these indexes and then another vector holding the numerical representation of the tags is created. Sentences are lengthened to be of the specific length and then tag vectors are also made to be of the same length as that

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sentence vectors to ensure consistency. In order to standardize the model, the tag vector is padded with the Other ('O') tag so that its length becomes the same as that of the word vector. This model makes use of an mBERT embedding vector with a dimension of 768 and a FastText embedding vector of 300 dimensions. The embedding vectors were then used to construct an embedding matrix for each embedding.

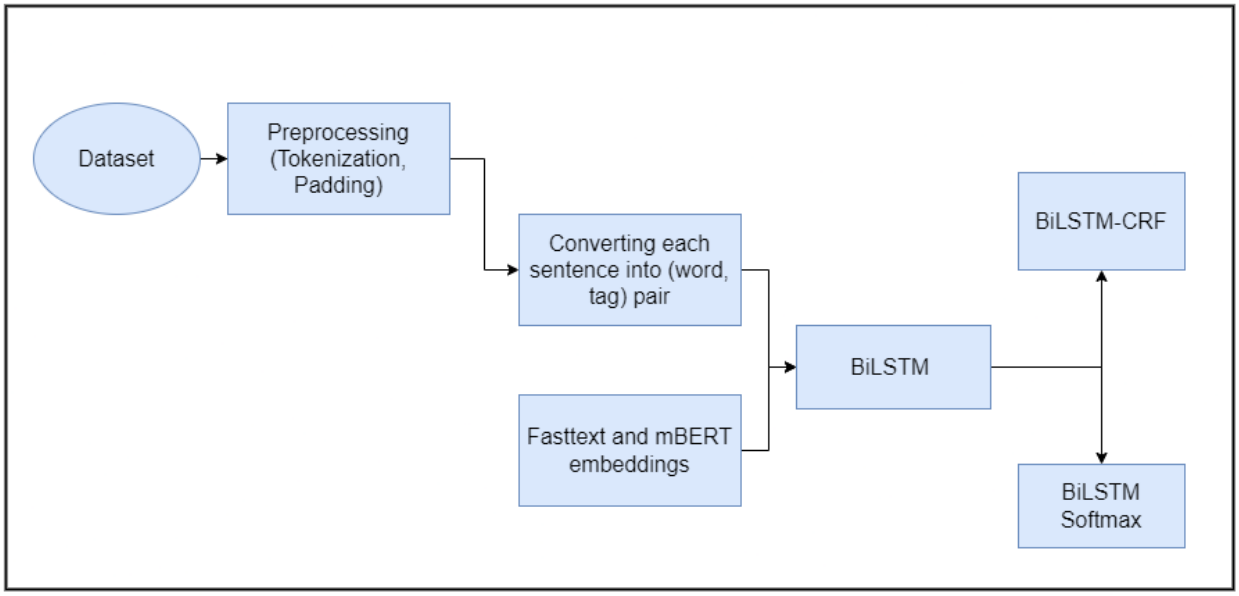

Fig. 3: Proposed methodology

D. Network Architecture

The paper presents biLSTM-CRF and biLSTM-Softmax models with FastText and mBERT embeddings. So a total of four models have been implemented namely BiLSTM CRF with FastText, BiLSTM CRF with mBERT, BiLSTM Softmax Classifier with FastText, and BiLSTM Softmax Classifier with mBERT. Out of four models, BiLSTM CRF with the Fasttext model outperformed everyone as seen in TABLE III.

TABLE III: Performance evaluation of all models

Model	F1-Score	Precision	Recall	Accuracy
Bi-LSTM CRF (FastText)	0.72	0.77	0.71	0.97
Bi-LSTM CRF (mBERT)	0.68	0.74	0.67	0.96
Bi-LSTM Softmax (FastText)	0.61	0.67	0.60	0.95
Bi-LSTM Softmax (mBERT)	0.64	0.71	0.62	0.96

A bi-directional LSTM combines two LSTMs, one of which moves forward (left to right) and another backward (left to right) capturing the entire meaning and context of the sentence. The context includes labels from the past and the future in a sequence. Using Bi-LSTM helps in considering both the past and the future information for NER. Undirected graphical models comprise conditional Random fields (CRFs). The conditional probability of values is computed using CRF. This probabilistic graph model considers the context of the surrounding areas while classifying data. Architecture for the same model is as follows:

The proposed deep learning model's starting point is represented by the input Layer, followed by the embedding layer. The pre-trained FastText model is loaded in order to obtain the word embeddings. This embedding layer output is fed to the two stacked Bidirectional LSTM layers. Both Bi-LSTM layers have a 0.1 recurrent dropout parameter. Then there is a Timedistributed layer with an activation function of ReLU, which is followed by a CRF layer. CRF layer having an output of dimension equal to the number of tags, is used to correctly classify them. The output dimension of 17 is used. Batch size used to train this model of seven layers is (including input and output layer) 32. The complete model is trained with the help of the RMSPROP optimizer. For each model, 50 epochs were picked.

IV. Results

The performance of the statistical or machine learning model is assessed using evaluation measures. Four different performance measures are taken into account to assess the model's performance [20] [21].

- Accuracy (A), which indicates how many data points can be correctly predicted.

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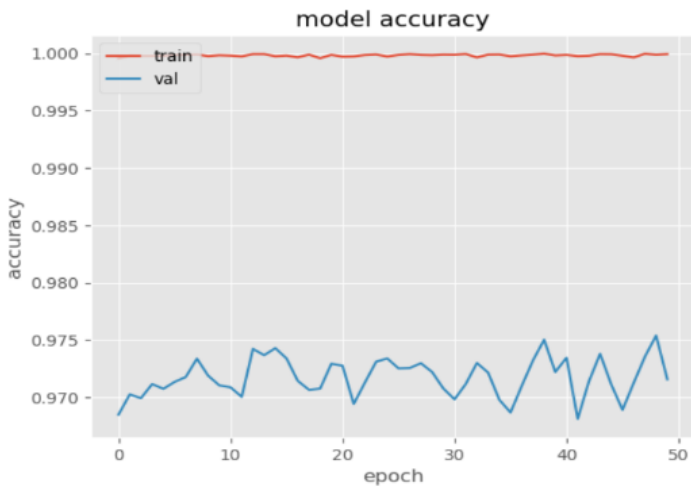

Fig. 4: Per epoch training and validation accuracy variation

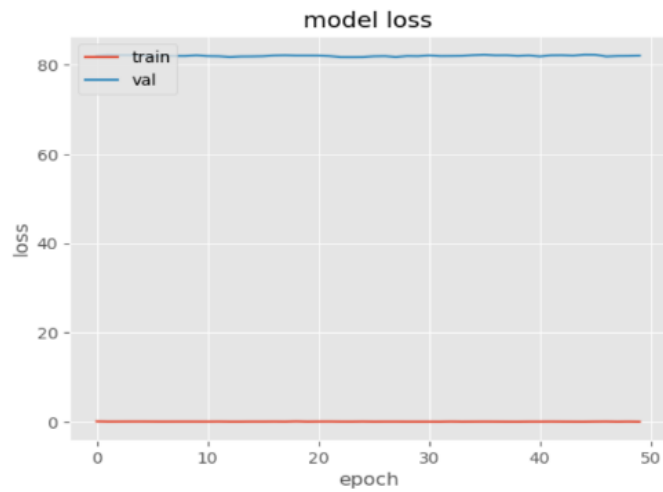

Fig. 5: Per epoch training and validation loss variation

$$Accuracy = \frac{Numberofcorrectpredictions}{Totalnumberofpredictions} \quad (1)$$

- A binary classifier statistic called precision (P) measures how accurately all positive labels are generated.

$$Precision = \frac{TP}{TP + FP} \quad (2)$$

Where TP is true positive and FP is false positive.

- Recall (R) measures the percentage of accurately predicted positive labels among all positive labels.

$$Recall = \frac{TP}{TP + FN} \quad (3)$$

Where TP is true positive and FN is false negative.

- F-measure signifies how many times the model made the correct prediction over the whole dataset. Mathematically it is the harmonic mean of recall (R) and precision (P).

$$F - measure = \frac{RP}{\frac{1}{2}(R + P)} \quad (4)$$

Various model architectures like Bi-LSTM CRF and Bi-LSTM Softmax, and word embeddings like FastText and mBERT have been tested in this study. Among all models, the Bi-LSTM CRF with fasttext embedding model reported the highest results for the experiment. TABLE IV shows tag-wise Precision, Recall, and f1-Score performance for the same model, on the weather dataset. Bi-LSTM CRF (FastText) gives the best performance with an f1-Score of 71.58%. Per epoch-wise accuracy and loss for both validation and training data are visualized in Fig 4 and Fig 5.

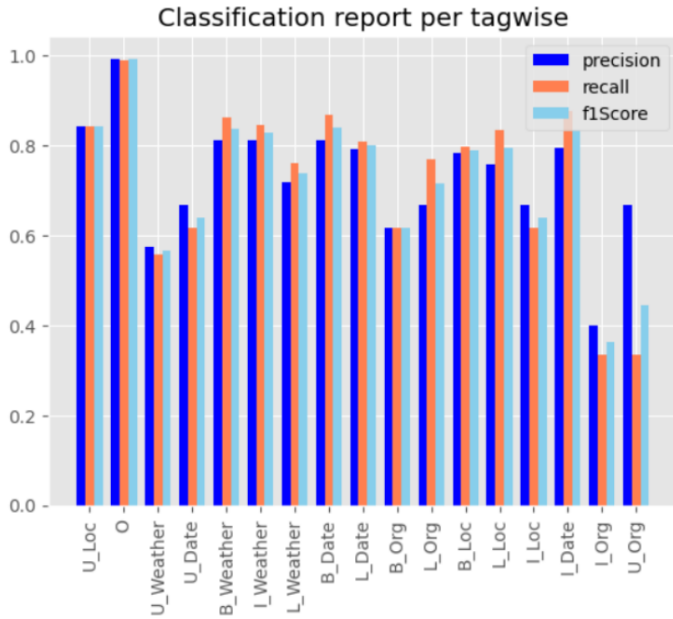

Fig. 6: Per tag performance evaluation

V. Conclusion and Future Scope

Named Entity Recognition (NER) is predominantly a domain-specific task. In contrast to other Indian languages, English literature has previously tackled the majority of Named Entity Recognition work. Since there were no annotated corpora available specifically for the weather domain, a sizable dataset of Marathi-named entities have been designed for NER work. The Bi-LSTM CRF model with FastText embedding had the best results out of the four models that have been discussed, with an accuracy rate of 97.45%. Since there has not previously been a NER system designed exclusively for the weather domain, the constructed dataset and suggested model contribute to novel work.

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TABLE IV: Tagwise Precision, Recall, and f1-Score

Tags	Precision	Recall	f1-Score
B_Weather	0.81	0.86	0.84
I_Weather	0.81	0.85	0.83
L_Weather	0.72	0.76	0.74
U_Weather	0.58	0.56	0.57
B_Date	0.81	0.87	0.84
I_Date	0.80	0.88	0.83
L_Date	0.79	0.81	0.80
U_Date	0.67	0.62	0.64
B_Loc	0.78	0.80	0.79
I_Loc	0.67	0.62	0.64
L_Loc	0.76	0.83	0.79
U_Loc	0.84	0.84	0.84
B_Org	0.62	0.62	0.62
I_Org	0.40	0.33	0.36
L_Org	0.67	0.77	0.71
U_Org	0.67	0.33	0.44
O	0.99	0.99	0.99

In the future, expanding the dataset size and using more expert tagging methods can enhance the performance. The data has been gathered from a single publication's website, and creating diversity in the weather domain dataset by differing the publications can help to create a more comprehensive model. Further, the model can be loaded into web servers to create a web application.

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